# Identifying the Causes of Protectionism using Double Machine Learning

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## 1 Proposed Topic

Why do some democracies protect domestic producers more than others? I aim to study whether factors such as exports, unemployment rates, GDP, inflation rates, and the number of political parties causally shape how protectionist a country's trade policy is. Using a panel of many countries spanning several decades, I will estimate the effect of each factor using double machine learning (DML), which lets flexible learners such as random forests and neural networks estimate causal parameters in the presence of complex confounding variables CITE double machine learning paper.

**Research Question:** *Which domestic factors cause countries to adopt more protectionist trade policy, and how robust are these effects?*

This analysis aims to build upon prior analyses such as Ehrlich 2007 and similar, CITE. Prior analyses have used linear regression to evaluate the causes of protectionism. I propose to use double machine learning to validate the results of Ehrlich 2007 and similar, CITE, creating a second avenue to test causal relationships in political science.

This analysis aims to test the following hypotheses: write a little more on why each is expected and why alternate would not be expected

> **H1 (Exports).** Exports being a higher percentage of GDP increases/decreases – which makes more sense? protectionism.

> **H2 (Unemployment).** Increased unemployment rates cause increased protectionism.

> **H3 (GDP).** Increased GDP per capita and total GPD in US$ I have data for both – combine or separate into two hypotheses? cause increased or decreased? protectionism.

> **H4 (Inflation).** Increased inflation rates cause increased protectionism.

> **H5 (Number of Political Parties).** More political parties causes increased protectionism

For the sake of this analysis, the level of protectionism will be quantified using tariff rates explain why this is a reasonable thing to do. In other words, higher tariff rates mean higher levels of protectionism, and vice versa.

## 2 Proposed Sources

### 2.1 Ehrlich 2007

Summarize methods and findings

### 2.2 Find similar to Ehrlich

### 2.3 Another similar paper to Ehrlich, find one

### 2.4 Double ML paper, need to locate this (ask Dr. MacDonald), also maybe python documentation

### 2.5 A paper or three on impacts of protectionism

## 3 Existing Analysis

More in depth summary of Ehrlich2007 and similar papers

## 4 Proposed Data and Methods

Data on tariff rates, exports, unemployment rates, GDP, and inflation CITE each dataset – wrote down link to each in google doc were obtained from World Bank Group CITE, and data on the number of political parties was obtained from Varieties of Democracy CITE both specific dataset and V-Dem.

The required data reduction and cleaning is complete. A simple Python program was written to pull the data needed from each dataset, and add the data to one large .csv file could maybe describe in more detail exactly what I did, though it was less painful here than for PHY466 project. Might be best to just link GitHub so anyone interested in what I did can find it there.

Make table (steal format from 466 proposal/paper) showing first few rows of dataset

Explain DML in detail, say how it works, which algorithms specifically, why those algorithms (likely random forest and neural networks)

## 5 Preliminary Results

## 6 Impact of Protectionism

# 7 Conclusion